

Electricity bill generator

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Introduction

Electricity is used in every home, and the electricity department charges customers according to the number of units consumed.

This project calculates the electricity bill automatically using conditional statements (if-else) and functions in C programming.

It helps reduce manual calculation errors and gives accurate results quickly.

objective

Objective 01

To calculate electricity bill automatically.

Objective 02

To understand if-else conditional statements.

Objective 03

To learn how functions improve code

Objective 04

To reduce manual calculation error

Problem Statement

Create a C program that calculates the electricity bill based on the following slab rates:

Units	Rate
0 – 100	₹1.5 per unit
101 – 200	₹2.5 per unit
Above 200	₹4 per unit

User stories :

01 Household Consumer

As a household electricity consumer, I want to enter my electricity units so that I can quickly know my monthly electricity bill without doing manual calculations.

02 Billing Operator

As a billing operator, I want the system to automatically apply the correct slab rates so that I can generate accurate electricity bills quickly and reduce calculation mistakes.

03 Electricity Consumer Monitoring Usage

As an electricity consumer, I want to calculate my bill based on different unit values so that I can understand how my electricity consumption affects my bill.

04 Electricity Department

As an electricity department, I want to calculate electricity bills automatically so that I can save time and avoid calculation mistakes.

Logic of program

Step 1

Take the Customer ID from the user.



Step 2

Take the Customer Name from the user.



Step 3

Take the number of electricity units consumed from the user.



Step 4

Pass the units to the calculateBill() function.



Step 5

Inside the function, use if-else statements to check the electricity slab.



Step 6

Calculate the bill according to the applicable slab rate.



Step 7

Return the calculated bill from the function.



Step 8

Display the Customer ID, Customer Name, Units Consumed, and Total Electricity Bill



Algorithm

- Start
- Read costumer Id
- Read costumer name
- Read units consumed.
- Call calculateBill().
- Check the slab:
- If units ≤ 100
- Else if units ≤ 200
- Else
- Calculate bill.
- Print bill.
- Stop.

```
#include <stdio.h>
```

```
// Function to calculate electricity bill
```

```
float calculateBill(int units)
```

```
{
```

```
    float bill;
```

```
    if (units <= 100)
```

```
    {
```

```
        bill = units * 1.5;
```

```
    }
```

```
    else if (units <= 200)
```

```
    {
```

```
        bill = (100 * 1.5) + ((units - 100) * 2.5);
```

```
    }
```

```
    else
```

```
    {
```

```
        bill = (100 * 1.5) + (100 * 2.5) + ((units - 200) * 4);
```

```
    }
```

```
    return bill;
```

```
}
```

```
int main()
```

```
{
```

```
    int customerID, units;
```

```
    char customerName[50];
```

```
    float bill;
```

```
    // Input customer details
```

```
    printf("Enter Customer ID: ");
```

```
    scanf("%d", &customerID);
```

```
    printf("Enter Customer Name: ");
```

```
    scanf(" %[^\n]", customerName);
```

```
// Input electricity units
printf("Enter Electricity Units Consumed: ");
scanf("%d", &units);

// Calculate bill
bill = calculateBill(units);

// Display bill
printf("\n=====\\n");
printf("          ELECTRICITY BILL\\n");
printf("=====\\n");
printf("Customer ID   : %d\\n", customerID);
printf("Customer Name : %s\\n", customerName);
printf("Units Consumed: %d\\n", units);
printf("Total Bill    : Rs. %.2f\\n", bill);
printf("=====\\n");

return 0;
```

Enter Customer ID: 108474804

Enter Customer Name: AMISHA

Enter Electricity Units Consumed: 75

=====

ELECTRICITY BILL

=====

Customer ID : 108474804

Customer Name : AMISHA

Units Consumed: 75

Total Bill : Rs. 112.50

=====

PS C:\Users\khush\Desktop\project21>

Enter Customer ID: 100959214

Enter Customer Name: Aashi

Enter Electricity Units Consumed: 150

=====

ELECTRICITY BILL

=====

Customer ID : 100959214

Customer Name : Aashi

Units Consumed: 150

Total Bill : Rs. 275.00

=====

PS C:\Users\khush\Desktop\project21> █

Enter Customer ID: 1074809210

Enter Customer Name: Bhumika

Enter Electricity Units Consumed: 250

=====

ELECTRICITY BILL

=====

Customer ID : 1074809210

Customer Name : Bhumika

Units Consumed: 250

Total Bill : Rs. 600.00

=====

PS C:\Users\khush\Desktop\project21> |

Future Improvement

- Add customer name and ID.
- Include GST and fixed charges.
- Save records in a file.
- Create a menu-driven program.
- Develop a graphical interface.

Conclusion

This project demonstrates how functions and conditional statements can be used together in C programming to solve a real-world problem. It calculates electricity bills accurately based on slab rates and provides a simple example of modular programming.

Thank You